Vladimir Borel

Problem 1 (Max Points 40)

Sport Zone's running shoes division in Zaragoza purchases a line of shoes at €30 from its supplier in China. As there is a long lead time, at the time of order placement, demand is still uncertain. Sport Zone has a forecast that its demand is normally distributed with mean 2,100 and standard deviation 600 for the upcoming season. In addition, Sport Zone recently found an alternative supplier in Portugal with a higher unit cost of €40 but with a negligible lead time. As the lead time from Portugal is very short, Sport Zone may make the order decision from Portugal even after demand is realized. Sport Zone sells the shoes at €50 each. The leftover inventory can be sold to discounters for €5.

- $p = 50, \quad c_c = 30, \quad c_p = 40, \quad s = 5$
- $D \sim \mathcal{N}(\mu = 2100, \sigma = 600)$

$$z = \frac{Q - \mu}{\sigma}, \quad L(z) = \varphi(z) - z(1 - \Phi(z))$$

$$E[\text{Short}] = \sigma L(z), \quad E[\text{Sold}] = \mu - E[\text{Short}], \quad E[\text{Left}] = Q - \mu + E[\text{Short}]$$

(a) If Sport Zone can only place an order from the supplier in China, how many shoes should Sport Zone buy to maximize its expected profit in the upcoming season? What is its expected profit? (5 points)

$$c_u = p - c_c = 50 - 30 = 20$$
$$c_o = c_c - s = 30 - 5 = 25$$
$$\text{cr} = \frac{c_u}{c_u + c_o} = \frac{20}{20 + 25} = 0.4444$$
$$z^* = \Phi^{-1}(\text{cr}) = -0.1397$$
$$Q^* = \mu + z^*\sigma = 2100 + (-0.1397)(600) = \boxed{2016.17} \approx 2016$$

$$L(z^*) = 0.4727$$
$$E[\text{Short}] = 600 \times 0.4727 = 283.61$$
$$E[\text{Sold}] = 2100 - 283.61 = 1816.39$$
$$E[\text{Left}] = 2016.17 - 2100 + 283.61 = 199.78$$

$$\begin{aligned}\pi_a &= pE[\text{Sold}] + sE[\text{Left}] - c_c Q \\ &= 50 \times 1816.39 + 5 \times 199.78 - 30 \times 2016.17 \\ &= \boxed{31333.17}\end{aligned}$$

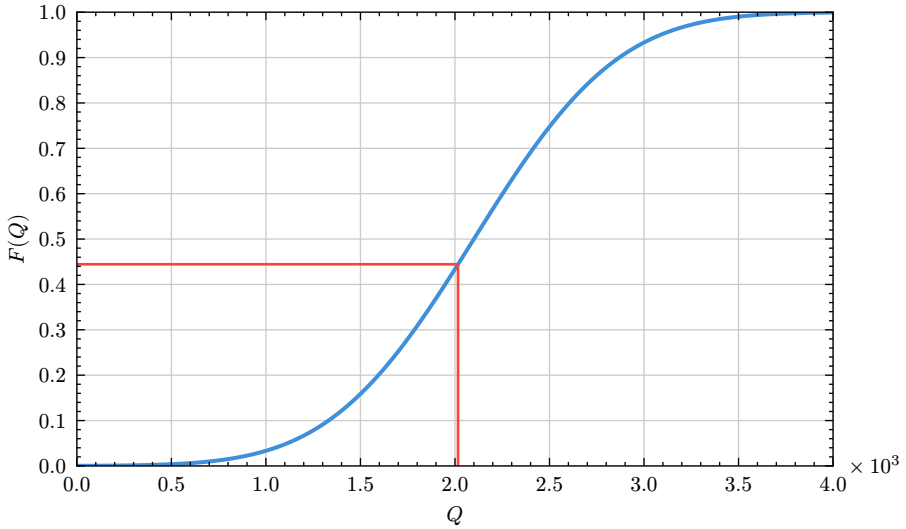


Figure 1: $Q^* = F^{-1}(\text{cr})$

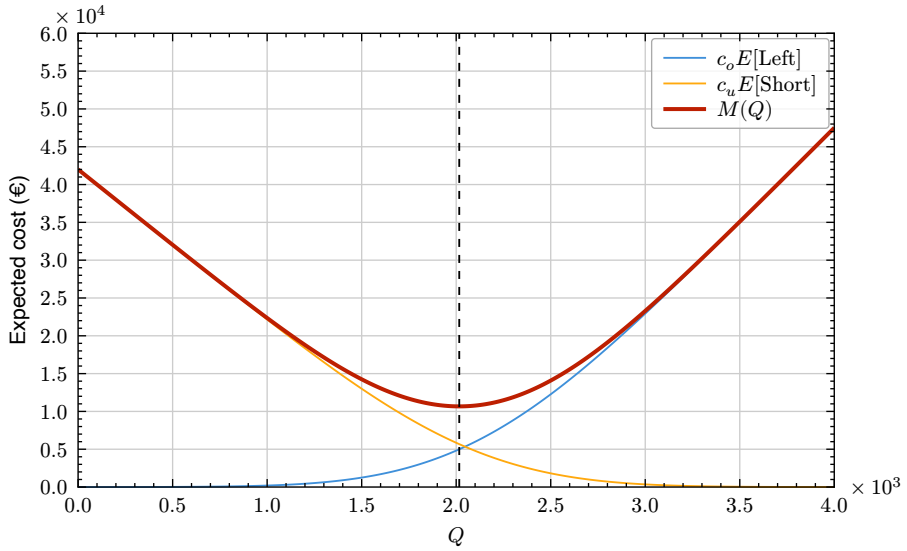


Figure 2: Mismatch cost $M(Q)$. $\pi(Q) = (p - c_c)\mu - M(Q)$, so Q^* minimizes M.

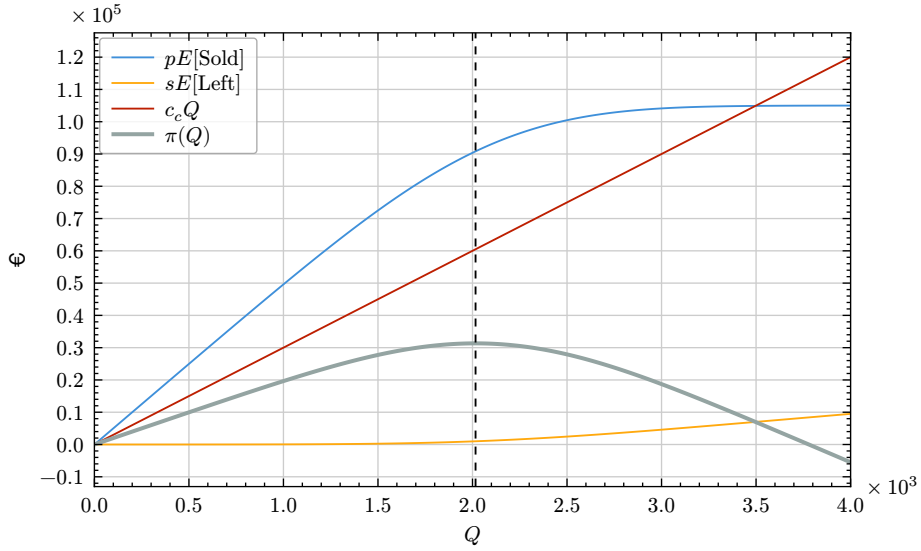


Figure 3: Components of expected profit.

(b) If Sport Zone can place an order from both suppliers, how many shoes should Sport Zone buy from China to maximize expected profit, especially considering that it still has a chance to order some quantity from Portugal after demand is realized? What is its expected profit in this case? (5 points)

- Demand above Q is ordered from Portugal after demand is known ⇒ no lost sales
- Underage = extra sourcing cost only

$$
\begin{aligned}
c_u &= (p - c_c) - (p - c_p) = c_p - c_c = 40 - 30 = 10 \\
c_o &= c_c - s = 30 - 5 = 25 \\
\text{cr} &= \frac{10}{10 + 25} = 0.2857 \\
z^* &= -0.5659 \\
Q^* &= 2100 + (-0.5659)(600) = \boxed{1760.43} \approx 1760
\end{aligned}
$$

$$
\begin{aligned}
E[\text{Portugal}] = E[\text{Short}] &= 600 \times 0.7442 = 446.49 \\
E[\text{Left}] &= 1760.43 - 2100 + 446.49 = 106.92
\end{aligned}
$$

- All μ units are sold at p:

$$
\begin{aligned}
\pi_b &= p\mu - c_c Q - c_p E[\text{Portugal}] + sE[\text{Left}] \\
&= 50 \times 2100 - 30 \times 1760.43 - 40 \times 446.49 + 5 \times 106.92 \\
&= \boxed{34861.98}
\end{aligned}
$$

- China order drops by $2016.17 - 1760.43 = 255.74$ units: a shortage now costs 10 € instead of 20 €

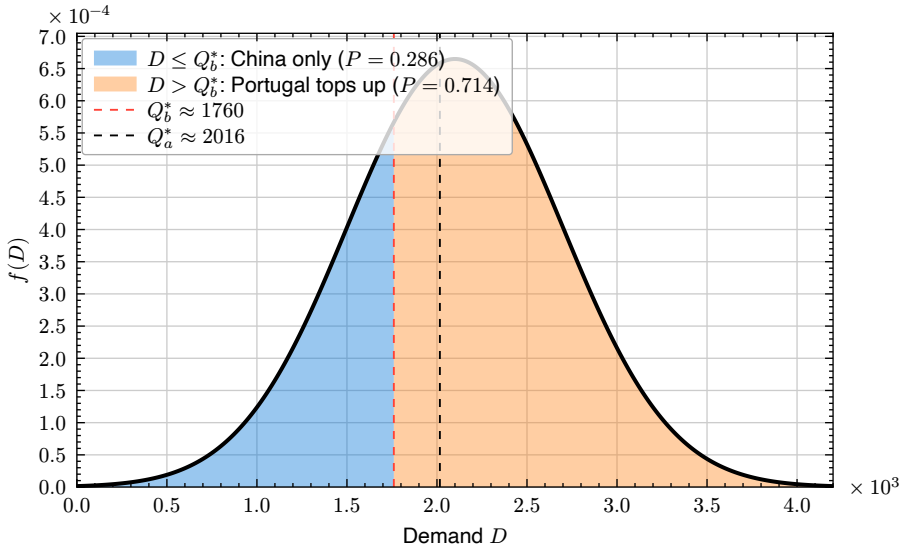


Figure 4: Demand split at the China order quantity.

(c) Now suppose that Sport Zone has to make a one-time contract payment to establish a business relationship with the Portuguese supplier. What is the maximum amount Sport Zone would be willing to pay to have this new supplier? (5 points)

- Max payment = expected profit gain from Portugal

$$\pi_b - \pi_a = 34861.98 - 31333.17 = \boxed{3528.81}$$

(d) Now back to (a), suppose that Sport Zone is considering a thesis project with ZLOG 2017 to improve their demand forecasting. If the company anticipates a more accurate forecast with the same mean but a reduced standard deviation of $(1-\theta) \times 600$ *(where* θ *is a number between 0 and 1), how much would Sport Zone be willing to pay for this project? Please express your response as a function of* θ *(rather than testing different values of* θ*). (5 points)*

- $z^* = \Phi^{-1}(\text{cr}) = -0.1397$ depends only on cr, not on σ
- $Q^* = \mu + z^*\sigma$, $E[\text{Short}] = \sigma L(z^*)$, $E[\text{Left}] = \sigma(z^* + L(z^*)) \Rightarrow$ all scale with σ

$$\begin{aligned}\pi(\sigma) &= (p - c_c)\mu - c_o E[\text{Left}] - c_u E[\text{Short}] \\ &= (p - c_c)\mu - \sigma[c_o z^* + (c_u + c_o)L(z^*)] \\ &= (p - c_c)\mu - (c_u + c_o)\varphi(z^*)\sigma\end{aligned}$$

- Last step uses $1 - \Phi(z^*) = c_o/(c_u + c_o)$
- Slope:

$$(c_u + c_o)\varphi(z^*) = (20 + 25) \times 0.3951 = 17.78 \text{ € per unit of } \sigma$$

- With $\sigma' = (1-\theta)\sigma$:

$$\begin{aligned}V(\theta) &= \pi((1-\theta)\sigma) - \pi(\sigma) \\ &= (c_u + c_o)\varphi(z^*)\sigma\theta \\ &= 17.78 \times 600 \times \theta \\ &= \boxed{10666.83\ \theta}\end{aligned}$$

- Check, $\theta = 1$: $V = 10666.83$ = mismatch cost in (a) $= (p - c_c)\mu - \pi_a = 10666.83$

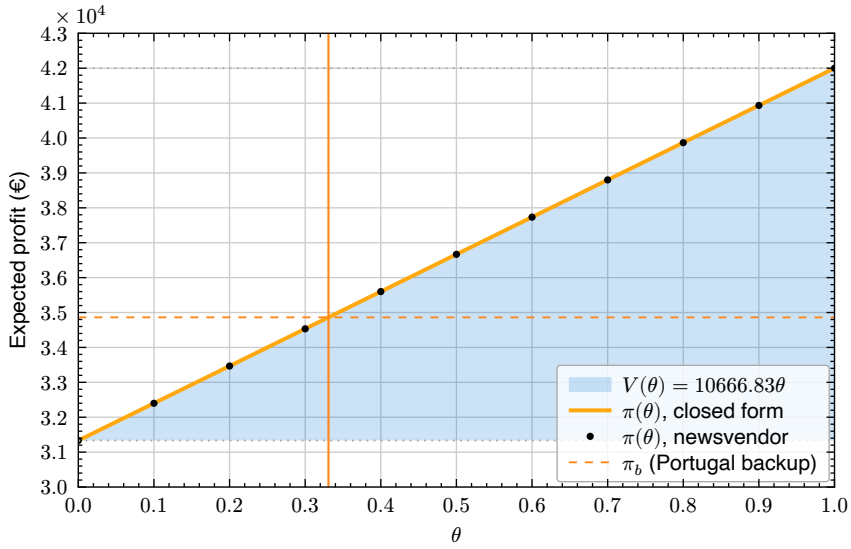


Figure 5: Project beats the Portugal backup for $\theta > 0.331$.

(e) What do you observe in (d) in terms of the impact of standard deviation on the expected profit? (5 points)

- $\pi^*(\sigma) = (p - c_c)\mu - 17.78\sigma$: linear, decreasing in σ
- Each unit of σ costs 17.78
- $\sigma = 0$: $\pi = (p - c_c)\mu = 42000$ (perfect information)
- $\sigma = 600$: $\pi = 31333.17$
- $z^* = -0.1397 < 0$ (cr < 0.5) $\Rightarrow Q^* < \mu$, gap grows with σ

(f) Back to (a), the company has realized from their history that if a customer can't find buy what he/she wants because of stock out, their reputation gets worse and they eventually lose some of the customers. They have estimated this undesirable impact as the monetary cost, which is €3 per each of lost sales. What is the optimal order quantity and the expected profit respectively? What do you observe here especially comparing this result with that of (a)? (5 points)

$$
\begin{aligned}
c_u &= p - c_c + g = 50 - 30 + 3 = 23 \\
c_o &= c_c - s = 25 \\
\text{cr} &= \frac{23}{23 + 25} = 0.4792 \\
z^* &= -0.0522 \\
Q^* &= 2100 + (-0.0522)(600) = \boxed{2068.65} \approx 2069
\end{aligned}
$$

$$
\begin{aligned}
\pi_f &= pE[\text{Sold}] + sE[\text{Left}] - gE[\text{Short}] - c_c Q \\
&= 50 \times 1844.63 + 5 \times 224.02 - 3 \times 255.37 - 30 \times 2068.65 \\
&= \boxed{30526.13}
\end{aligned}
$$

- vs (a): Q^* up by 52.48, π down by 807.04
- Goodwill raises $c_u \Rightarrow$ higher cr $\Rightarrow$ order more
- Profit falls: each stockout now costs g on top of the lost margin

(g) In (f), what are the optimal order quantity and the expected profit if the monetary cost per each of lost sales is €10, €20, €30 and €40? What do you observe here especially comparing this result with that of (f)? (5 points)

g	cr	Q^*	$Q^* - \mu$	π^*
0	0.4444	2016.17	–83.83	31333.17
3	0.4792	2068.65	–31.35	30526.13
10	0.5455	2168.51	68.51	28920.45
20	0.6154	2276.03	176.03	27096.64
30	0.6667	2358.44	258.44	25638.01
40	0.7059	2424.84	324.84	24427.54

- Higher $g \Rightarrow$ higher cr $\Rightarrow$ higher Q^*
- $Q^* > \mu$ once cr $> 0.5 \Leftrightarrow c_u > c_o \Leftrightarrow g > c_o - (p - c_c) = 5$
- Expected profit keeps falling as g rises
- Expensive stockouts $\Rightarrow$ order well above mean demand

(h) In (a), suppose Sport Zone has to pay a fixed transportation cost of €500 whenever it places an order in addition to other costs. How many shoes should Sport Zone buy to maximize its expected profit in the upcoming season? What is its expected profit? (5 points)

- Fixed cost does not change c_u or c_o $\Rightarrow$ same $Q^* = 2016.17$
- Still order, since $\pi_a - 500 > 0$

$$\pi_h = \pi_a - 500 = 31333.17 - 500 = \boxed{30833.17}$$

Problem 2 (Max Points 35)

Sam Holding (SH) sells a tent to Intersport (IP). SH's variable cost per tent is €100 while its wholesale price is €185 per tent (the wholesale price includes the cost of shipping the tent to IP). IP sells the tent for €250. Suppose IP's forecast for season sales can be described with a Poisson distribution with rate (mean) 8.75. Furthermore, IP plans to only buy once from SH and any eventual leftovers are sold at a discount of 75% (i.e., at €62.5 per tent).

- $c = 100, \quad w = 185, \quad p = 250, \quad s = 62.5$
- $D \sim \text{Poisson}(\lambda = 8.75)$

$$E[\text{Short}] = \lambda(1 - F(Q-1)) - Q(1 - F(Q))$$
$$E[\text{Sold}] = \lambda - E[\text{Short}], \quad E[\text{Left}] = Q - E[\text{Sold}]$$

(a) How many tents should IP order? (5 points)

$$c_u = p - w = 65$$
$$c_o = w - s = 122.5$$
$$\text{cr} = \frac{c_u}{c_u + c_o} = 0.3467$$

$$F(6) = 0.2305 < 0.3467 \leq F(7) = 0.354 \Longrightarrow Q^* = \boxed{7}$$

(b) What is IP's expected profit given his optimal order quantity in (a)? (5 points)

$$E[\text{Short}] = 2.21$$
$$E[\text{Sold}] = 8.75 - 2.21 = 6.54$$
$$E[\text{Left}] = 7 - 6.54 = 0.46$$

$$\begin{aligned}\pi_{\text{IP}} &= pE[\text{Sold}] + sE[\text{Left}] - wQ \\ &= 250 \times 6.54 + 62.5 \times 0.46 - 185 \times 7 \\ &= \boxed{368.59}\end{aligned}$$

(c) What is SH's expected profit given IP's order in part (a)? (5 points)

$$\pi_{\text{SH}} = (w - c)Q = (185 - 100) \times 7 = \boxed{595}$$

(d) To maximize the supply chain's total profit (SH's profit plus IP's profit) how many tents should be shipped to IP? (5 points)

- Supply chain buys at c, not w

$$c_u = p - c = 150$$
$$c_o = c - s = 37.5$$
$$\text{cr} = 0.8$$

$$F(10) = 0.7352 < 0.8 \leq F(11) = 0.8266 \Longrightarrow Q^* = \boxed{11}$$

(e)–(f): Suppose SH were to accept unsold tents at the end of the season. IP would incur a €15 shipping cost per tent returned to SH. Suppose SH gives IP a 90% credit for each returned tent, that is, SH pays IP €166.5 for each returned tent.

- $b = 0.9w = 166.5$
- IP's net value per leftover: $b - \text{shipping} = 151.5$

(e) How many tents should IP order to maximize its profit? (5 points)

$$c_u = p - w = 65$$
$$c_o = w - (b - \text{shipping}) = 33.5$$
$$\text{cr} = 0.6599$$

$$F(9) = 0.6203 < 0.6599 \leq F(10) = 0.7352 \Longrightarrow Q^* = \boxed{10}$$

(f) What is the expected profit for IP and SH, given IP's order in part (e)? (5 points)

$$E[\text{Short}] = 0.67$$
$$E[\text{Sold}] = 8.75 - 0.67 = 8.08$$
$$E[\text{Left}] = 10 - 8.08 = 1.92$$

$$\begin{aligned}\pi_{\text{IP}} &= pE[\text{Sold}] + (b - \text{shipping})E[\text{Left}] - wQ \\ &= 250 \times 8.08 + 151.5 \times 1.92 - 185 \times 10 \\ &= \boxed{460.47}\end{aligned}$$

$$\begin{aligned}\pi_{\text{SH}} &= (w - c)Q - bE[\text{Left}] \\ &= (185 - 100) \times 10 - 166.5 \times 1.92 \\ &= \boxed{529.62}\end{aligned}$$

- Assumes returned tents are worth nothing to SH

(g) To maximize the supply chain's total profit, what should IP's buyback percentage be? (The current credit is 90%). Assuming that SH can sell all returned tents at €185 per tent, is profit maximizing buyback percentage feasible? (5 points)

- Coordination: IP's cr = supply chain's cr from (d)

$$\frac{p - w}{p - w + w - (b - \text{shipping})} = \frac{p - w}{p - b + \text{shipping}} = 0.8$$

$$b^* = p + \text{shipping} - \frac{p - w}{\text{cr}_{\text{SC}}} = 250 + 15 - \frac{65}{0.8} = \boxed{183.75}$$

$$\frac{b^*}{w} = \frac{183.75}{185} = \boxed{99.32\ \%}$$

- IP then orders $Q = 11$ = supply-chain optimum from (d)
- SH pays $b^* = 183.75$, resells at $r = 185 \Rightarrow$ keeps $r - b^* = 1.25$ per return $\geq 0 \Rightarrow$ **feasible**

$$\pi_{\text{IP}} = pE[\text{Sold}] + (b^* - \text{shipping})E[\text{Left}] - wQ = 498.93$$
$$\pi_{\text{SH}} = (w - c)Q - (b^* - r)E[\text{Left}] = 938.32$$

- vs 90% credit (f): IP 498.93 vs 460.47 SH 938.32 vs 885.6 (with resale) $\Rightarrow$ both better off

Problem 3 (Max Points 15)

Each year the admissions committee at a top business school receives a large number of applications for admission to the MBA program and they have to decide on the number of offers to make. Since some of the admitted students may decide to pursue other opportunities, the committee typically admits more students than the ideal class size of 720 students. You were asked to help the admissions committee estimate the appropriate number of people who should be admitted. It is estimated that in the coming year the number of people who will not accept the admission offer is normally distributed with mean 50 and standard deviation 21. Suppose for now that this school does not maintain a waiting list, that is, all students are accepted or rejected.

- $R \sim \mathcal{N}(\mu = 50, \sigma = 21)$: admitted students who decline
- Admit $Q \Rightarrow$ class size $Q - R$

(a) Suppose 750 students are admitted. What is the probability that the class size will be at least 720 students? (5 points)

$$P(Q - R \geq 720) = P(R \leq 750 - 720) = P(R \leq 30)$$
$$z = \frac{30 - 50}{21} = -0.9524$$
$$P(R \leq 30) = \Phi(-0.9524) = \boxed{0.1705}$$

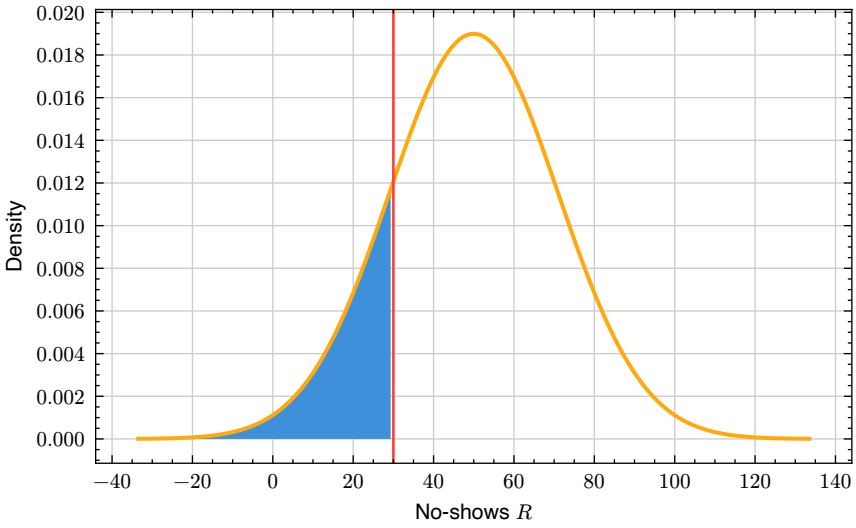


Figure 6: Shaded: $P(R \leq 30)$ = class has at least 720 students.

(b) There is a mutual agreement that it is about two times more expensive to have a student in excess of the ideal 720 than to have fewer students in the class. What is the appropriate number of students to admit? (5 points)

- Newsvendor on buffer $y = Q - 720$
- $R < y \Rightarrow$ class too big (overage); $R > y \Rightarrow$ too small (underage)

$$\text{cr} = \frac{c_u}{c_u + c_o} = \frac{1}{1+2} = 0.3333$$
$$z^* = \Phi^{-1}(0.3333) = -0.4307$$
$$y^* = \mu + z^*\sigma = 50 + (-0.4307)(21) = 40.95$$
$$Q^* = 720 + y^* = 760.95 \approx \boxed{761}$$

(c) A waiting list mitigates the problem of having too few students since at the very last moment there is an opportunity to admit some students from the waiting list. Hence, the admissions committee revises its estimate: It claims that it is five times more expensive to have a student in excess of 720 than to have fewer students accept among the initial group of admitted students. What is your revised suggestion? (5 points)

$$\text{cr} = \frac{1}{1+5} = 0.1667$$
$$z^* = \Phi^{-1}(0.1667) = -0.9674$$
$$y^* = 50 + (-0.9674)(21) = 29.68$$
$$Q^* = 720 + y^* = 749.68 \approx \boxed{750}$$

- Waiting list makes shortage cheap $\Rightarrow$ lower cr $\Rightarrow$ admit fewer (761 $\rightarrow$ 750)

Problem 4 (Max Points 10)

In the formal presentation of the Newsvendor problem, we have shown that the profit maximization problem is equivalent to the minimizing of the following cost function:

$$C(Q) = E[C_u \cdot (D-Q)^+ + C_o \cdot (Q-D)^+] = C_u E[\text{Short}] + C_o E[\text{Left}]$$

Assume that the demand is normally distributed with mean μ and standard deviation σ. By using the characterization (i.e., expression) for the optimal order quantity, Q^*, show that the optimal cost can be expressed as $C(Q^*) = A(C_u, C_o) \times \sigma$, where $A(C_u, C_o)$ is a constant which is a function of the cost parameters C_u and C_o but not of the parameters (μ, σ) of the demand distribution D. What can you infer for the maximum profit that the newsvendor can gain by ordering the optimal quantity?

Optimal quantity

- $D = \mu + \sigma Z$, $Z \sim \mathcal{N}(0,1)$

$$\Phi(z^*) = \frac{C_u}{C_u + C_o}, \quad Q^* = \mu + z^*\sigma$$

- z^* depends only on C_u, C_o

Short and left

- $L(z) = \varphi(z) - z(1 - \Phi(z))$
- $(Q-D)^+ = (Q-D) + (D-Q)^+$

$$E[\text{Short}] = \sigma E\left[(Z - z^*)^+\right] = \sigma L(z^*)$$
$$E[\text{Left}] = (Q^* - \mu) + E[\text{Short}] = \sigma(z^* + L(z^*))$$

Optimal cost

$$\begin{aligned} C(Q^*) &= C_u \sigma L(z^*) + C_o \sigma(z^* + L(z^*)) \\ &= \sigma[(C_u + C_o)L(z^*) + C_o z^*] \end{aligned}$$

- Use $1 - \Phi(z^*) = C_o/(C_u + C_o)$:

$$\begin{aligned} (C_u + C_o)L(z^*) &= (C_u + C_o)\varphi(z^*) - z^*(C_u + C_o) \cdot \frac{C_o}{C_u + C_o} \\ &= (C_u + C_o)\varphi(z^*) - C_o z^* \end{aligned}$$

- $C_o z^*$ cancels:

$$C(Q^*) = (C_u + C_o)\varphi(z^*) \cdot \sigma$$

$$\boxed{A(C_u, C_o) = (C_u + C_o)\,\varphi\left(\Phi^{-1}\left(\frac{C_u}{C_u + C_o}\right)\right)}$$

Maximum profit

- $C_u = p - c$, $C_o = c - s$
- $E[\text{Sold}] = \mu - E[\text{Short}]$, $Q = E[\text{Sold}] + E[\text{Left}]$

$$\begin{aligned} \Pi(Q) &= pE[\text{Sold}] + sE[\text{Left}] - cQ \\ &= (p - c)\mu - (p - c)E[\text{Short}] - (c - s)E[\text{Left}] \\ &= C_u \mu - C(Q) \end{aligned}$$

$$\boxed{\Pi(Q^*) = C_u \mu - A(C_u, C_o)\sigma}$$

Takeaways

- $C_u \mu$: profit with perfect information; $A\sigma$: cost of uncertainty
- Max profit falls linearly in σ; halve $\sigma \Rightarrow$ halve mismatch cost
- $\Pi(Q^*)/(C_u \mu) = 1 - (A/C_u) \cdot \sigma/\mu \Rightarrow$ coefficient of variation σ/μ is what matters
- Profitable only if $\sigma/\mu < C_u/A$

Quiz Problem (Max Points 20)

On a given Boston-Barcelona flight, there are 200 seats. Suppose the ticket price is $475 on average and the number of passengers who reserve a seat but do not show up for departure is normally distributed with mean 30 and standard deviation 15. You decide to overbook the flight and estimate that the average loss from a passenger who will have to be bumped (if the number of passengers exceeds the number of seats) is $800.

- $N \sim \mathcal{N}(\mu = 30, \sigma = 15)$: no-shows
- Y = reservations above 200
- $N < Y \Rightarrow$ bumped (overage); $N > Y \Rightarrow$ empty seat (underage)

(a) What is the maximum number of reservations that should be accepted? (2 points)

$$c_u = 475, \quad c_o = 800$$
$$\text{cr} = \frac{475}{475 + 800} = 0.3725$$
$$z^* = \Phi^{-1}(0.3725) = -0.3251$$
$$Y^* = \mu + z^*\sigma = 30 + (-0.3251)(15) = 25.12$$
$$\text{Reservations} = 200 + Y^* = 225.12 \approx \boxed{225}$$

(b) Suppose you allow 220 reservations. How much money do you expect to pay out in compensation to bumped passengers? (2 points)

- Bumped $= (Y - N)^+$ = "leftover"

$$z = \frac{Y - \mu}{\sigma} = \frac{20 - 30}{15} = -0.6667, \quad L(z) = 0.8178$$
$$E[(N - Y)^+] = \sigma L(z) = 12.27$$
$$E[\text{Bumped}] = (Y - \mu) + \sigma L(z) = -10 + 12.27 = 2.27$$
$$E[\text{Payout}] = 800 \times 2.27 = \boxed{\$1813.44}$$

(c) Suppose you allow 220 reservations. What is the probability that you will have to deal with bumped passengers? (2 points)

$$P(N < 20) = \Phi\left(\frac{20 - 30}{15}\right) = \Phi(-0.6667) = \boxed{0.2525}$$

(d)–(g): Now assume there is no overbooking, and the high fare is $675 and the low fare is $375. Demand for the low fare is abundant while demand for the high fare is normally distributed with a mean of 80 and standard deviation 35.

- $D_H \sim \mathcal{N}(\mu = 80, \sigma = 35)$: high-fare demand
- Q = protection level; low fare gets $200 - Q$ seats, always sold

(d) What is the probability of selling 200 reservations if you set an optimal protection level (i.e., number of seats reserved) for the full fare? (8 points)

$$c_u = r_H - r_L = 300, \quad c_o = r_L = 375$$
$$\text{cr} = \frac{300}{300 + 375} = 0.4444$$
$$z^* = \Phi^{-1}(0.4444) = -0.1397$$
$$Q^* = 80 + (-0.1397)(35) = 75.11 \approx 75$$

- Booking limit for low fare: $200 - 75 = 125$, always filled
- All 200 sold $\Leftrightarrow D_H \geq Q^*$

$$P(D_H \geq 75) = 1 - \Phi\left(\frac{75 - 80}{35}\right) = 1 - \Phi(-0.1429) = \boxed{0.5568}$$

- With unrounded Q^*: $1 - \text{cr} = 0.5556$

(e) Suppose a protection level of 85 is established. What is the average number of lost high-fare passengers? (2 points)

$$z = \frac{85 - 80}{35} = 0.1429, \quad L(z) = 0.3316$$
$$E[\text{Lost}] = \sigma L(z) = 35 \times 0.3316 = \boxed{11.61}$$

(f) Continue to assume a protection level of 85 is established. What is the expected number of unoccupied seats? (2 points)

- Only protected seats can fly empty

$$E[\text{Empty}] = E\left[(Q - D_H)^+\right] = (Q - \mu) + E[\text{Lost}] = 5 + 11.61 = \boxed{16.61}$$

(g) Again, assume a protection level of 85 is established. What is the expected revenue from the flight? (2 points)

$$E[\text{High sold}] = \mu - E[\text{Lost}] = 80 - 11.61 = 68.39$$
$$\text{Low sold} = 200 - 85 = 115$$
$$E[\text{Revenue}] = 675 \times 68.39 + 375 \times 115 = \boxed{\$89291.48}$$